A-A-54187
25 June 1990

COMMERCIAL ITEM DESCRIPTION

FREEZER, LABORATORY

(Ultra Low Temperature, High Capacity)

The General Services Administration has authorized the use of this Interim Commercial Item Description.

This Commercial Item Description covers ultra-low temperature biological or blood plasma storage freezers.

Salient characteristics:

Style and Design. The freezer shall be a chest style, ultra-low temperature, biological freezer capable of storing in bulk (without racks) a minimum of 700 units of red cells in plastic bags and individually packaged in corrugated cardboard boxes. The corrugated cardboard boxes, supplied separately from the freezer, shall measure 7" long X 5-1/2" wide X 1-3/4" high. It should be noted, however, that boxes expand after filling and cooling. Therefore, capacity required shall be for 700 filled and cooled boxes which measure 7-1/8 inches by 5-5/8 inches by 1-7/8 inches. The boxes shall be specifically designed for the primary bag freezing method.

The overall external dimensions (LxWxH) of the freezer, including all hardware and controller, shall not exceed 110.0 cubic feet.

The freezer shall operate at an operator selected temperature range of at least -85°F to -121°F.

Pull-down capability of an empty chamber shall be from an initial temperature of 75°F to -121°F in 3 hours maximum without aid of back-up or additional cooling system.

AMSC N/A FSC 4110

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The refrigeration system shall be configured to have the dual capability for either air cooling, water cooling, or a combination of air and water cooling, and have sufficient reserve capacity to maintain an internal temperature of -121°F or colder when the ambient air temperature is maintained at 95°F plus or minus 2°F and the ambient temperature of the compressor cooling water is maintained at 85°F plus or minus 2°F for a minimum of 96 hours.

Maximum current draw shall not increase during this 96 hour test period. Operation in an ambient temperature of less than 90°F shall permit the compressors to cycle on and off while maintaining the -121°F internal temperature.

The unit shall be equipped with an external temperature indicating device and a seven day continuous recording chart, single pen, inkless, pressure writing system. The temperature indicator shall run on battery power if the line power fails. Steady-State temperature variation shall not exceed plus or minus 6°F from the readings at the sensing and control bulbs.

Construction. The interior and exterior of the freezer shall be fabricated of stainless or corrosion protected steel plate having a minimum thickness of 22 gage. Between the interior and exterior walls and bottom, there shall be poured or foamed in-place rigid urethane insulation all around having a minimum thickness of 5 inches plus or minus 3/16 inches and 3-1/2 inches plus or minus 3/16 inches in lid. The freezer shall be supplied with a magnetically sealed door lid gasket constructed with a material which remains flexible and usable at the expected temperatures in use. The freezer shall have a minimum of three movable or removable sublids constructed of Lexan or equal. Sublids should allow access to part of the contents without exposing the entire contents to ambient conditions when the freezer is opened. The freezer lid shall be counter balanced or strutted to hold any position from closed to fully opened with a maximum movement force of ten pounds in either direction.

The freezer shall be equipped with a dual cascade refrigeration system consisting of two independent cascade refrigeration systems. Each cascade refrigeration system shall consist of two separate compressor/motor combinations for high and low stage cooling. Each of the four compressor/motor combinations shall be rated at not less than 1 horse power. The compressor shall operate at a nominal speed of 1750 RPM and be of the accessible or "semi" hermetic type. Each cascade system shall have an on/off switch.

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a. Under normal condition of operation to maintain internal temperature at a pre-set level, the cascade refrigeration systems shall alternate from one system to the other.

b. For rapid cool down of the contents of the freezer, means shall be provided to allow automatic simultaneous operation of both refigeration systems.

Such operation will be effected automatically when the internal temperature of the freezer exceeds a predetermined set point. The temperature set point shall be variable over a range of 6-9°F warmer than -121°F.

c. Failure of either refrigeration system shall cause the remaining system to function and maintain the pre-set temperature. In addition, a warning system, visual or audible or both, shall alert or signal the failure.

Anti-Icing. The unit shall be supplied with anti-icing to the extent that it will prevent freezer lock of the freezer door lid or of the contents of the freezer when the unit is operated in a 95°F plus or minus 2°F, 95 percent relative humidity plus or minus 5 percent at sea level atmosphere. Opening the lid fully for 60 plus or minus 2 seconds every 4 hours for 24 hours shall produce no frost build-up or freeze-lock of door or contents with a full, half-full, or empty freezer. Formation of non-adhering ice crystals is acceptable.

Safety and Alarm Systems. The freezer shall be supplied with cam actuated locking mechanisms with one padlock and keys. The locking mechanisms shall be located on the front and center on the door lid and freezer. The lock mechanisms shall permit key removal in either locked or unlocked position. The freezer shall have a line and battery-powered, (back-up) audible and visual alarm system for power failure and temperature deviations. The alarm must be adjustable to plus or minus 2°F within the operating range. The alarm system must activate the audible and visual alarms when the temperature exceeds the preset low or high limits. The alarm must be able to be silenced temporarily with a push button or similar device, or defeated with a key switch. The alarm must include contacts for connection to a remote alarm and be compatible with a computerized central alarm/monitor system. The alarm system must continue to function for a minimum of 5 hours when AC power is lost to the system.

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Electrical characteristics. The freezer shall operate over the full tolerance range of one or more of the following line voltage and frequency combinations as specified by the customer requisition.

1. 220 VAC plus or minus 10 percent, 50 Hz plus or minus 2 Hz, three phase

2. 208 VAC plus or minus 10 percent, 60 Hz plus or minus 2 Hz, three phase

Label. The freezer shall have a readily accessible permanently mounted label with at least the following information:

1. Item Identification

2. Manufacturer

3. Contract Number and Date of Manufacture

4. Unit's Serial Number

5. Electrical Requirements

6. Maximum Sustained Operating Current Draw

7. Locked Compressor Rotor Current Draw

8. Refrigerant Type and Amount Required

9. Maximum Current draw of Battery Charger

The above information may be provided on two labels provided they are mounted adjacent to one another.

Underwriters Laboratories Inc. The freezer shall conform to UL 471-Standard for Safety Refrigeration Units. Acceptable evidence of this shall be a UL listing mark or a certified test report from an independent laboratory acceptable to the Government.

Optional Equipment. When specified by the customer requisition, the following equipment shall be supplied with the freezer:

a. Remote alarm indicators
b. Standby, automatically activated voltage booster to protect freezer in case of reduced voltage conditions (brown-out).

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Operators' and Service Data Manuals.

The contractor shall furnish manuals, handbooks and/or brochures containing complete operation, installation and service/maintenance instructions (including pictures of illustrations, as necessary) with complete schematics and wiring diagrams. All manuals, handbooks, and/or brochures shall be written in the English language and all schematics and wiring diagrams shall use American electrical and electronics symbols. The manuals shall include electrical data and connection diagrams for all applicable utilities. The instructions shall also contain a complete list of all replaceable parts showing part number, name and quantity required. All service notes, service memos, etc., pertaining to the equipment and issued subsequent to the printing of the original instructions shall also be included. These instructions shall be, as a minimum, that furnished to service engineers (or servicemen) that normally install and service the equipment for the company or distributor. When the system being procured includes, or shall operate ancillary equipment, the service data shall include complete instructions and drawings which show interfacing of all system components.

Two complete sets of the above information shall be packaged with each unit of the required equipment. Where multiple quantities of the same unit are to be delivered to the same ultimate consignee, then only a total of two service manuals need to be supplied to each consignee. However, two operational manuals still need to be supplied with each unit.

Finish. The freezer shall be fabricated from materials which are inherently resistant to corrosion or which are treated to render them corrosion resistant. Where painting or plating is used, appropriate cleaning and priming shall be performed prior to finishing. Finish shall provide a pleasing professional appearance and be free of gaps, scratches, chips and other imperfections.

Workmanship. The freezer shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One freezer, as specified, constitutes one unit.

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QUALITY ASSURANCE PROVISIONS.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

First article inspection. First article inspection shall be performed on the freezer when a first article sample is required. This inspection shall include examination and test.

Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the item identification, the name of the component/material, the lot number and the lot size.

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Sampling for examination and testing. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 1 (percent defective) and an inspection level of II. The unit of product for sampling purposes shall be one freezer unless the requisition calls for a quantity of both 50 and 60 Hz units. In this case, the freezers equipped with a 50 Hz system shall be considered a different lot from the 60 Hz freezers and inspected accordingly.

Tests. Tests shall be conducted to determine compliance with this specification.

Metric products.

Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerance specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and

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regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Numbers are covered by this document.

4110-01-306-5093
4110-01-320-5621

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD - MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 4110-0438

Location:
CID/FREEZER/LAB/H1B-22

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-54187	FREEZER, LABORATORY (Ultra Low Temperature, High Capacity)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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